

Spring Semester, 2021, Faculty of Architecture, CEPT University

Course:

AD4600: Theory and Practice of Details

Course Tutor/s:

Sankalpa

Link to CEPT Portfolio :

(studio work from past semesters)

Long Description :

In continuously changing construction environment in India; exposure to the world has brought in significant amount of new materials, construction system and details. This has radically affected not only the way we build but also the way we detail. Whereas a large part of the country is trying to deal with influences in various ways; each region based on the available skills, tools, material and technology are adapting to those influences resulting in a patchwork of details. In such a situation, dealing with influences of all kinds within a given context of artisanal knowledge and available technology to accomplish it is a challenge that need to be addressed by re emphasizing the theory and practice behind the details so that the fundamental principles and its application in deriving it is understood clearly instead of direct accepting of a detail. The course will trace down the morphological and typological attributes of detail and share some of the standard ways of analyzing and detailing. Students will have to undertake case studies as well as site visit in order to fulfill the basic requirements of this course.

Course Expense :

(INR -extra expenditure a student would incur during the semester)

INR 2000/- (Two Thousand Rupees)

Course Prerequisite:

Only for 2nd semester of MAD students.

Learning Outcomes :

After completing the Courses,the student will be able to :

- To compare and analyze a building on the basis of its detail (using morphological principle)
- To analyze a detail on the basis of its structural, constructional and performative role.
- To apply 'morphological method' of detailing
- To recognize the difference in detailing between compatible and incompatible materials (material properties) or morphology
- To detail a building as an integration of choices of design idea, constructability, process, cost, environment, use of resources, time, performance and function.

Weekly Plan &
Assessment:

Weeks		Assessment	
		%	Criteria
Week 1	Details (Dimensions of Details)	0	
Week 2	Principles of Detailing	0	
Week 3	Development of detailing	0	
Week 4	Development of detailing	0	
Week 5	Joining compatible and incompatible material	0	
Week 6	Junction & Joinery	50	Clarity of concept, ability to effectively communicate & produce technically correct drawing
Week 7	Analyzing a Detail	0	
Week 8	Analyzing a Detail	0	
Week 9	Detailing as an Integral part of design	0	
Week 10	Attitude to details: Presentation by practicing architects	0	
Week 11	Field Visit	0	
Week 12	Summing Up & Final Submission	50	Demonstrate clarity of concept, ability to effectively communicate & produce technically correct drawing

References :

1. Allen, Edward, and Joseph Iano. Fundamentals of Building Construction: Materials and Methods. Sixth ed.
2. Allen, Edward. Architectural Detailing: Function Constructibility Aesthetics. New York: Wiley, 1993.
3. Deplazes, Andrea. Constructing Architecture: Materials, Processes, Structures: A Handbook. 3. Extended ed. Basel: Birkhauser, 2013.
4. Gambhir, M. L., and Neha Jamwal. Building Materials: Products, Properties and Systems. New Delhi: Tata McGraw Hill, 2011.
5. Illston, J. M. Construction Materials Their Nature and Behaviour. 3rd ed. London: Spon, 2001.
6. Jagadish, K. S. Building with Stabilized Mud. New Delhi: I.K. International Pub. House, 2007.
7. Mehta, Madan, and Walter Scarborough. Building Construction: Principles, Materials, and Systems. Upper Saddle River, N.J.: Pearson/Prentice Hall, 2008.
8. Messler, Robert W. Joining of Materials and Structures from Pragmatic Process to Enabling Technology. Amsterdam: Elsevier, 2004.
9. Minke, Gernot. Building with Bamboo. Basel: Birkhauser, 2012.
10. Wachsmann, Konrad. The Turning Point of Building; Structure and Design. New York: Reinhold Pub., 1961.
11. Wahl, Iver. Building Anatomy: An Illustrated Guide to How Structures Work. New York: McGrawHill, 2007.
12. N, Gyula, and Chris Pollington. New Architecture and Technology. Oxford, UK: Architectural, 2003.

Tutor Profile:

Sankalpa is an architect and urban designer. He has been teaching construction, structure and design studio since several years in the Faculty of Architecture, CEPT University UG program. His design studios involve methods of teaching construction that are bodily involved; modelling techniques that can be experienced and verified. He is one of the founding member of architectural practices 'thumbimpressions collaborative'. Research and development on 'bamboo as a building material' is his current area of interest. He has guided several students on thesis related to material, construction and urban design. He has been also actively involved in community based initiative related to construction. He has been also actively involved in community based initiative related to construction.

